

Motivated and capable embedded software engineer, eager to apply multi-disciplinary problem solving to complex tasks and projects. Passionate about empowering and educating others through technology and art.

Core Competencies

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|-------------|---------|-------------------------|
| • C++ | • Rust | • Software Architecture |
| • Python | • Yocto | • Technical Writing |
| • Linux/GNU | • CAD | • Graphic Design |

Principal Embedded Software Engineer - BAE Systems SMS

Apr 2020 - Present

- Developed C++ microservice application suite for phased antenna array systems
- Built and tested Linux images for custom hardware using Yocto
- Developed Rust MQTT microservices for satellite payload reuse baseline
- Created Python GUI library for embedded system control
- Wrote abstraction layers for 3D render engines for use in capability simulation frameworks
- Conducted phone screens and reviewed resumes for potential hire candidates
- Led working groups for software development and provided career guidance

Staff Physicist I - Stable Laser Systems

Nov 2014 - Jan 2020

- Developed bare-metal firmware in C++ for real-time control of mixed-signal laser system electronics
- Designed microcontroller boards implementing UART, SPI, and I2C interfaces
- Experience with physically-constrained PCB design using CAD models
- Designed, assembled, and tested electronics and performance of ultrastable laser systems
- Created scripts to streamline workflows and automate data collection and processing
- Designed and manufactured custom electronics enclosures using SolidWorks

B.S. Electrical and Computer Engineering - Rice University

Aug 2010 - Jun 2014

- Completed graduate-level courses in signal processing and filter design
- Gained lab experience in circuit design, embedded systems programming, and FPGAs
- Developed a bluetooth-connected, 3D-printed, panoramic (OpenCV) microscope for Android smartphones
- Maintained a balanced academic experience with Spanish, Chinese, and on-campus leadership

Personal Projects

Video Game Modding

- Custodian of community development for 2000+ Rocket League modders
- Scripted and recorded 10 hours of video tutorial content for Skillquest.io
- Wrote and hosted a Vuepress-based website with 50,000 words of tutorial content

Hidden Layer Design

- Started an art-meets-technology project with engineer friends
- Brought to life an LED-illuminated, laser cut PCB art piece which has been displayed in several galleries
- Created an interactive music art installation for the Museum of Boulder

Blender 3D

- Wrote a publicly-available extension to import Kerbal Space Program designs into Blender
- Taught a Rice University accredited semester-long course
- Gave an internal training course to 20 engineers at Ball Aerospace